

## Tutorial 2

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### Selection of Aggregate Structure

- For a 12.5 million ESA, select a 19 mm Superpave aggregate Structure based on the following data

|           | Blend 1 | Blend 2 | Blend 3 |
|-----------|---------|---------|---------|
| $G_{mb}$  | 2.457   | 2.441   | 2.477   |
| $G_{mm}$  | 2.598   | 2.558   | 2.664   |
| $G_b$     | 1.025   | 1.025   | 1.025   |
| $P_b$     | 5.9     | 5.7     | 6.2     |
| $P_s$     | 94.1    | 94.3    | 93.8    |
| $P_d$     | 4.5     | 4.5     | 4.5     |
| $G_{sb}$  | 2.692   | 2.688   | 2.665   |
| $H_{ini}$ | 125     | 133     | 125     |
| $H_{des}$ | 115     | 118     | 115     |

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# Formulae

- Adjusting volumetric properties to VTM=4.0%

$$P_{be} = P_{bt} - 0.4 * (4 - VTM_t)$$

$$VMA_{est} = VMA_t + C * (4 - VTM_t)$$

$$VFB_{est} = 100 \frac{(VMA_{est} - 4.0)}{VMA_{est}}$$

$$\%Gmm_{Nini_{est}} = \%Gmm_{Nini} - (4 - VTM_t)$$

$$P_{be_{est}} = P_{b,est} - \frac{P_s * P_{ba}}{100}$$

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# Superpave Design Criteria

|                                                   |         |         |        |       |     |
|---------------------------------------------------|---------|---------|--------|-------|-----|
| VTM                                               | 4.0%    |         |        |       |     |
| D/B                                               | 0.6-1.2 |         |        |       |     |
| %TSR                                              | 80.0%   |         |        |       |     |
| % VMA requirements for different Max Nominal size |         |         |        |       |     |
| Size                                              | 37.5    | 25      | 19     | 12.5  | 9.5 |
| %VMA                                              | 11%     | 12%     | 13%    | 14%   | 15% |
| % G <sub>mm</sub> Requirements at                 |         |         |        |       |     |
| ESA in Millions                                   | Nini ≤  | N Des = | Nmax ≤ | VFB   |     |
| <0.3                                              | 91.5    | 96      | 98     | 70-80 |     |
| 0.3-3                                             | 90.5    | 96      | 98     | 65-78 |     |
| 3-10                                              | 89.0    | 96      | 98     | 65-75 |     |
| 10-30                                             | 89.0    | 96      | 98     | 65-75 |     |

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## Un-adsuted Properties

|                  | Blend1 | Blend2 | Blend3 |
|------------------|--------|--------|--------|
| P <sub>b</sub>   | 5.9    | 5.7    | 6.2    |
| G <sub>se</sub>  | 2.875  | 2.812  | 2.979  |
| P <sub>ba</sub>  | 2.42   | 1.68   | 4.05   |
| %G <sub>mm</sub> | 94.573 | 95.426 | 92.980 |
| %VTM             | 5.4    | 4.6    | 7.0    |
| %VMA             | 14.1   | 14.4   | 12.8   |
| %VFB             | 61.5   | 68.2   | 45.2   |
| %Gmm @Nini       | 87.007 | 84.664 | 85.542 |

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## Adjusted Properties at VTM 4.0%

|           | Bld #1 | Bld#2 | Bld#3 | Criteria |
|-----------|--------|-------|-------|----------|
| Pest      | 6.47   | 5.93  | 7.41  |          |
| VMAest    | 13.83  | 14.25 | 12.21 | >13      |
| VFBest    | 71.08  | 71.93 | 67.25 | 65-75    |
| %Gmm Nini | 88.4   | 85.2  | 88.6  | <89      |
| Pbeest    | 4.2    | 4.3   | 3.7   |          |
| D/Best    | 1.07   | 1.04  | 1.23  | 0.6-1.2  |

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# Superpave Mix Design

If the max Nominal Size is 19 mm and the design traffic is  $20 \times 10^6$  ESA, Determine the optimum Binder Content

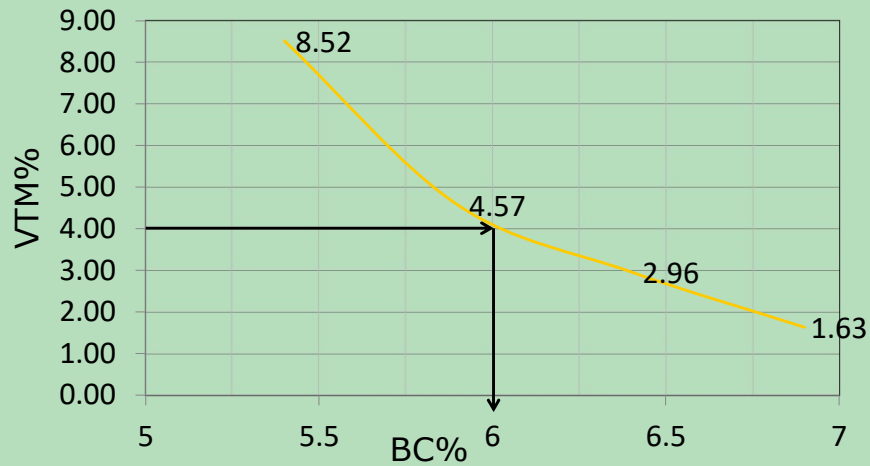
| $P_b$     | 5.4   | 5.9   | 6.4   | 6.9   |
|-----------|-------|-------|-------|-------|
| $G_{mb}$  | 2.351 | 2.441 | 2.455 | 2.469 |
| $G_{mm}$  | 2.57  | 2.558 | 2.53  | 2.51  |
| $G_b$     | 1.025 | 1.025 | 1.025 | 1.025 |
| $P_s$     | 94.6  | 94.1  | 93.6  | 93.1  |
| $P_d$     | 4.5   | 4.5   | 4.5   | 4.5   |
| $G_{sb}$  | 2.688 | 2.688 | 2.688 | 2.688 |
| $H_{ini}$ | 125   | 131   | 126   | 130   |
| $H_{des}$ | 115   | 118   | 114   | 112   |

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| $P_b$          | 5.4  | 5.9  | 6.4  | 6.9  |
|----------------|------|------|------|------|
| $G_{se}$       | 2.81 | 2.82 | 2.81 | 2.81 |
| $P_{ba}$       | 1.7  | 1.8  | 1.7  | 1.7  |
| $P_{be}$       | 3.8  | 4.2  | 4.8  | 5.3  |
| D/B            | 1.2  | 1.1  | 0.9  | 0.8  |
| VTM            | 8.52 | 4.57 | 2.96 | 1.63 |
| VMA            | 17.3 | 14.5 | 14.5 | 14.5 |
| VFB            | 50.6 | 68.6 | 79.6 | 88.7 |
| $\%G_{mmNini}$ | 84.2 | 86.0 | 87.8 | 84.7 |

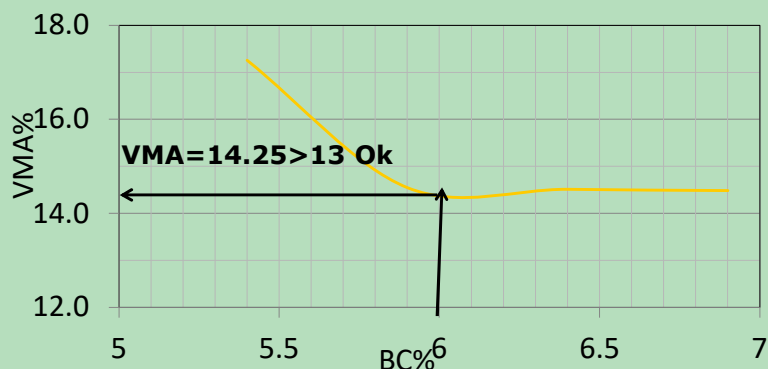
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# Determination of Design Binder Content



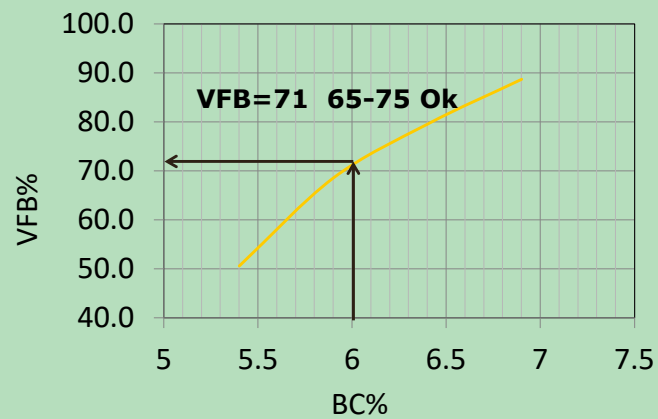
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# Check VMA Against Superpave Criteria



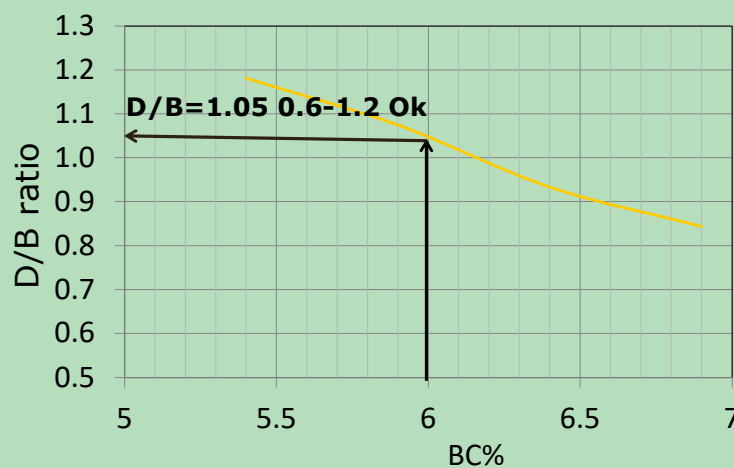
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## Check VFB Against Criteria



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## Check D/B Ratio Against Criteria



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